

Trajectory Simplification for Large-Scale Movement Data

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A) Introduction

Trajectory data records how objects move over time, such as vehicles in a city, ships at sea, and people during sports activities. Modern sensing systems collect trajectories at high sampling rates, which produces very dense sequences of points. This density increases storage costs, slows down analysis, and makes many downstream tasks harder, including indexing, similarity search, pattern mining, and visualization.

Trajectory simplification addresses these issues by compressing trajectories while preserving the essential geometric shape and movement characteristics. The key challenge is the trade-off between compression and quality. A method should remove redundant points aggressively, but it must keep critical points that affect error, shape, and query outcomes. Because different applications care about different quality notions, an effective simplification approach needs clear error measures, controllable compression, and strong empirical evidence across datasets.

B) Detailed Objectives

- Study the trajectory simplification problem on public datasets such as Microsoft GeoLife¹ and characterize key data properties (sampling irregularity, noise, speed changes, and trajectory length distribution) that influence simplification quality.
- Implement and compare representative baselines (e.g., classical and learning-based methods) under unified settings. Report results across multiple compression ratios and multiple quality measures (e.g., geometric error and task-oriented metrics).
- Propose a new trajectory simplification method to handle a concrete weakness of existing approaches, such as unstable error under irregular sampling and noise. The method targets a fixed compression budget while keeping key points around turns, stops, and speed changes.
- Evaluate efficiency and scalability, including runtime, memory usage, and throughput on large-scale trajectories, and identify which design choices dominate performance.

C) References

- Wang, Z., Long, C., Cong, G., & Jensen, C. S. (2024). *Collectively simplifying trajectories in a database: A query accuracy driven approach*. ICDE, 4383–4395. IEEE.
- Wang, Z., Long, C., & Cong, G. (2021). *Trajectory simplification with reinforcement learning*. ICDE, 684–695. IEEE.

¹<https://www.microsoft.com/en-us/research/publication/geolife-gps-trajectory-dataset-user-guide/>

- Zhang, D., Ding, M., Yang, D., Liu, Y., Fan, J., & Shen, H. T. (2018). *Trajectory simplification: an experimental study and quality analysis*. PVLDB, 11(9), 934–946.